

APPLICATION NOTE

lifelines-hc: a Python package for Higher Criticism testing in two-sample survival analysis under non-proportional hazards

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Abstract

Motivation: The log-rank test is the standard for comparing two survival distributions but is optimally powered only under proportional hazards (PH). Modern biological settings — immune checkpoint immunotherapy, aging-genetics models, and hormone-responsive pharmacology — increasingly produce temporally concentrated hazard departures. Existing weighted log-rank alternatives (Gehan-Wilcoxon, Tarone-Ware, Peto-Prentice, Fleming-Harrington) pre-specify a temporal emphasis, rendering them insensitive to departures outside their focal window.

Results: We introduce **lifelines-hc**, a Python package extending the **lifelines** survival library with the Higher Criticism (HC) test for right-censored two-sample data. HC evaluates the collective extremity of per-interval hypergeometric p-values and detects any localised hazard departure without committing to a temporal weight. Across three biological case studies — an immuno-oncology trial (CheckMate 057 PFS, $n = 582$), a simulated *Drosophila* late-life quantitative trait locus ($n = 1600$), and an adjuvant bisphosphonate trial (AZURE DFS, $n = 3359$) — HC achieves $p \leq 0.013$ while the log-rank test and all four weighted alternatives return $p \geq 0.10$.

Availability and Implementation: **lifelines-hc** is freely available under the MIT licence at <https://github.com/TODO/lifelines-hc> and via `pip install lifelines-hc`; it requires Python ≥ 3.8 and **lifelines** ≥ 0.29 . Full analysis scripts and per-method results tables are provided as Supplementary Data.

Key words: survival analysis; higher criticism; non-proportional hazards; hypothesis testing; Python; immuno-oncology

Introduction

The Kaplan-Meier estimator [Kaplan and Meier, 1958] and the log-rank test [Mantel, 1966] remain the workhorses of two-sample survival analysis in biomedical research. The log-rank statistic sums weighted observed-minus-expected event counts across all failure times and is asymptotically optimal when the hazard ratio between groups is constant throughout follow-up — the proportional hazards (PH) assumption [Schoenfeld, 1981].

Modern biological and clinical data increasingly violate PH in structured ways. Immune checkpoint inhibitors require an immune-priming phase before generating durable tumour suppression, producing survival curves that cross before diverging. This pattern is characteristic of the second-line NSCLC trial CheckMate 057, in which nivolumab showed an early progression excess relative to docetaxel before the immunotherapy arm gained a durable advantage [Borghaei et al., 2015]. In aging genetics, mutation-accumulation theory [Medawar, 1952] predicts alleles whose deleterious effects are expressed exclusively in post-reproductive life, creating narrow late-life mortality hot-spots that global averaging dilutes. In pharmacology, benefits conditional on a patient subgroup

generate composite hazard differences that partially cancel: in the AZURE trial of adjuvant zoledronic acid, a bisphosphonate, the drug's bone-microenvironment benefit requires the low-oestrogen state of postmenopause and was absent in the many premenopausal patients enrolled [Coleman et al., 2011].

Weighted log-rank tests were developed to address non-PH alternatives. Gehan-Wilcoxon [Gehan, 1965] and Peto-Prentice [Peto and Peto, 1972] down-weight late events to increase sensitivity to early hazard differences; Tarone-Ware [Tarone and Ware, 1977] applies an intermediate weight; Fleming-Harrington [Fleming and Harrington, 1991] provides a configurable weight via parameters (p, q) . However, each test concentrates power on a specific temporal region, so a test optimised for early differences is correspondingly insensitive to late ones, and no single pre-specified weight simultaneously covers all non-PH patterns.

Higher Criticism (HC; Donoho and Jin 2004) is a meta-test that detects non-random concentrations of small p-values within a collection of tests. Applied to survival analysis [Kipnis, 2021], HC divides the follow-up into equal-width intervals, derives a per-interval hypergeometric p-value comparing event rates

between the two groups, and assesses whether the most deviant subset of these p-values is collectively implausible under the global null. Because HC does not pre-weight any particular time window it retains power for concentrated hazard departures wherever they occur along the follow-up axis.

Despite this theoretical advantage, an accessible Python implementation integrated with standard survival analysis workflows has been lacking. We present **lifelines-hc**, which fills this gap by extending the widely-used **lifelines** library [Davidson-Pilon, 2019].

The lifelines-hc package

Higher Criticism test for survival data

Given two groups with right-censored observations (T_i, E_i, G_i) , partition the follow-up range into K equal-width intervals. For interval k , let n_k be the total number of events, r_k^A and r_k^B the subjects at risk in groups A and B, and x_k the number of events in group B. Under the null of equal group-specific hazards,

$$x_k \mid n_k, r_k^A, r_k^B \sim \text{Hypergeometric}(r_k^A + r_k^B, r_k^B, n_k), \quad (1)$$

yielding a one-sided interval p-value p_k . Ordering the K p-values as $p_{(1)} \leq \dots \leq p_{(K)}$, the HC statistic is

$$HC_K = \max_{1 \leq j \leq \lfloor \gamma K \rfloor} \frac{j/K - p_{(j)}}{\sqrt{p_{(j)}(1 - p_{(j)})/K}}, \quad (2)$$

where $\gamma \in (0, 1)$ (default 0.2) restricts scanning to the most extreme fraction of interval p-values [Donoho and Jin, 2004]. A global p-value is obtained by permuting group labels ($n_{\text{perms}} = 500$ by default). For the two-sided alternative, the minimum of the two one-sided interval p-values replaces p_k in (2).

Software design and API

lifelines-hc exposes a function-based API consistent with **lifelines.statistics**, accepting NumPy arrays of event times and indicators:

```
from lifelines_hc import (higher_criticism_test,
                        fisher_combination_test,
                        min_p_test,
                        suspected_deviations)

result = higher_criticism_test(
    T_A, T_B,
    event_observed_A=E_A, event_observed_B=E_B,
    n_intervals_to_pool=100, n_permutations=500,
    alternative='both', gamma=0.2, seed=42)

print(result.p_value)           # permutation-
                                # calibrated p-value
print(result.test_statistic)    # HC statistic
                                # value
```

The companion **suspected_deviations()** function returns a **pandas.DataFrame** of per-interval p-values with a **suspected** boolean column marking HC-flagged intervals, enabling overlay of diagnostic shading on Kaplan-Meier plots. Two complementary omnibus tests are also provided: Fisher combination [Fisher, 1932], which aggregates evidence via $-2 \sum_k \log p_k$, and a Bonferroni-corrected MinP test. All three share the same permutation infrastructure. The package

additionally ships with plotting utilities (**plot_km_with_hc**, **plot_pvalue_profile**) and a convenience wrapper **run_all_tests()** that benchmarks HC, Fisher, MinP, log-rank, and the four weighted alternatives in a single call, returning a **pandas.DataFrame** for easy comparison.

Results

We demonstrate **lifelines-hc** on three biological datasets with distinct non-PH structures (Figure 1), benchmarking HC against the log-rank test and four weighted alternatives. Full per-method p-value tables and the *Drosophila* case study are provided in Supplementary Data.

Immuno-oncology: crossing survival curves

CheckMate 057 [Borghaei et al., 2015] randomised 582 patients with advanced non-squamous NSCLC to nivolumab ($n = 292$) or docetaxel ($n = 290$). Progression-free survival showed crossing curves: docetaxel provided a short-term advantage during immune priming, subsequently overtaken by the durable nivolumab response (Figure 1A). Individual patient data were reconstructed from the published Kaplan-Meier curve via the algorithm of Guyot et al. [2012] as implemented in the **kmdata** R package.

The log-rank test is non-significant ($p = 0.351$) as the early excess and late benefit partially cancel. Among the weighted alternatives, Gehan-Wilcoxon ($p = 0.041$) and Peto-Prentice ($p = 0.049$) reach borderline significance by detecting the early crossing, but Tarone-Ware ($p = 0.358$) and Fleming-Harrington(1,1) ($p = 0.256$) do not. HC achieves $p = 0.002$ and characterises the full crossing structure with far greater power (Figure 1B) by flagging both the early divergence and late separation windows independently.

Aging genetics: a narrow late-life mortality hot-spot

To represent late-life QTL biology in *Drosophila melanogaster* [Medawar, 1952, Nuzhdin et al., 2005], we generated a synthetic piecewise-exponential dataset ($n = 800$ per genotype) with baseline median lifespan 60 days, 90-day censoring, and a $4\times$ elevated hazard exclusively in days 60–64 (the QTL window), with identical hazards outside this window. With 75 flanking null intervals, the log-rank test is non-significant ($p = 0.104$). All four weighted alternatives also fail: early-weighting tests (Gehan-Wilcoxon and Peto-Prentice, both $p = 0.203$) are insensitive to the late signal, and Fleming-Harrington(1,1), which emphasises mid-follow-up, reaches only $p = 0.060$. HC ($p = 0.002$) identifies the concentrated spike in the interval p-value profile (Supplementary Figure S1), demonstrating that its scanning approach is not penalised by the long null early period.

Adjuvant bisphosphonate therapy: a menopause-dependent effect

The AZURE trial [Coleman et al., 2011, 2014] randomised 3359 early-stage breast cancer patients to standard adjuvant therapy with ($n = 1681$) or without ($n = 1678$) zoledronic acid. Disease-free survival showed no significant log-rank difference ($p = 0.305$) because the drug’s bone-microenvironment benefit depends on the low-oestrogen state of postmenopause, which was absent in the many premenopausal patients enrolled at baseline. As these patients progressively transitioned to postmenopause during the decade-long follow-up, the hazard difference accumulated

in a specific mid-to-late window (months 20–60) rather than uniformly (Figure 1C–D).

All four weighted alternatives are non-significant ($p \geq 0.28$): no single temporal weight captures a pattern distributed across this window while remaining unaffected by the null early period. HC ($p = 0.012$) detects this scattered but collectively significant signal. The result is validated by published subgroup analyses showing significant DFS benefit in postmenopausal patients [Coleman et al., 2011, 2014], confirming that HC is detecting a genuine biological effect.

Across the three case studies, the two borderline significances for Gehan-Wilcoxon and Peto-Prentice in CheckMate 057 are the only instances where a weighted alternative reaches $p < 0.05$, and these are attributable to those tests' early emphasis coincidentally aligning with the crossing-curve signal rather than to any general superiority of the weighted approach. Fisher combination ($p \leq 0.04$ in all three domains) and MinP ($p \leq 0.022$ in two of three) are also significant, consistent with the sparse, localised nature of the underlying signals.

Conclusion

lifelines-hc provides a general-purpose HC test for two-sample right-censored survival data within the Python ecosystem. Unlike weighted log-rank alternatives, HC requires no a priori specification of the temporal window in which a hazard departure is expected and retains power for concentrated departures wherever they occur. The package integrates transparently with **lifelines** workflows, returns diagnostic interval-level information identifying which time windows drive the test result, and includes Fisher combination and MinP tests as complementary omnibus alternatives. Together, these tools offer a practical complement to standard survival testing in any setting where the proportional hazards assumption is uncertain.

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Conflict of Interest

None declared.

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Data availability

Individual patient data for CheckMate 057 and the AZURE trial were reconstructed from published Kaplan-Meier curves using the algorithm of Guyot et al. [2012] as implemented in the **kmdata** R package (<https://github.com/raredd/kmdata>). The synthetic *Drosophila* dataset is generated by the reproducible analysis scripts distributed with the package. All analysis code is available at <https://github.com/TOD0/lifelines-hc/tree/main/AppNote>.

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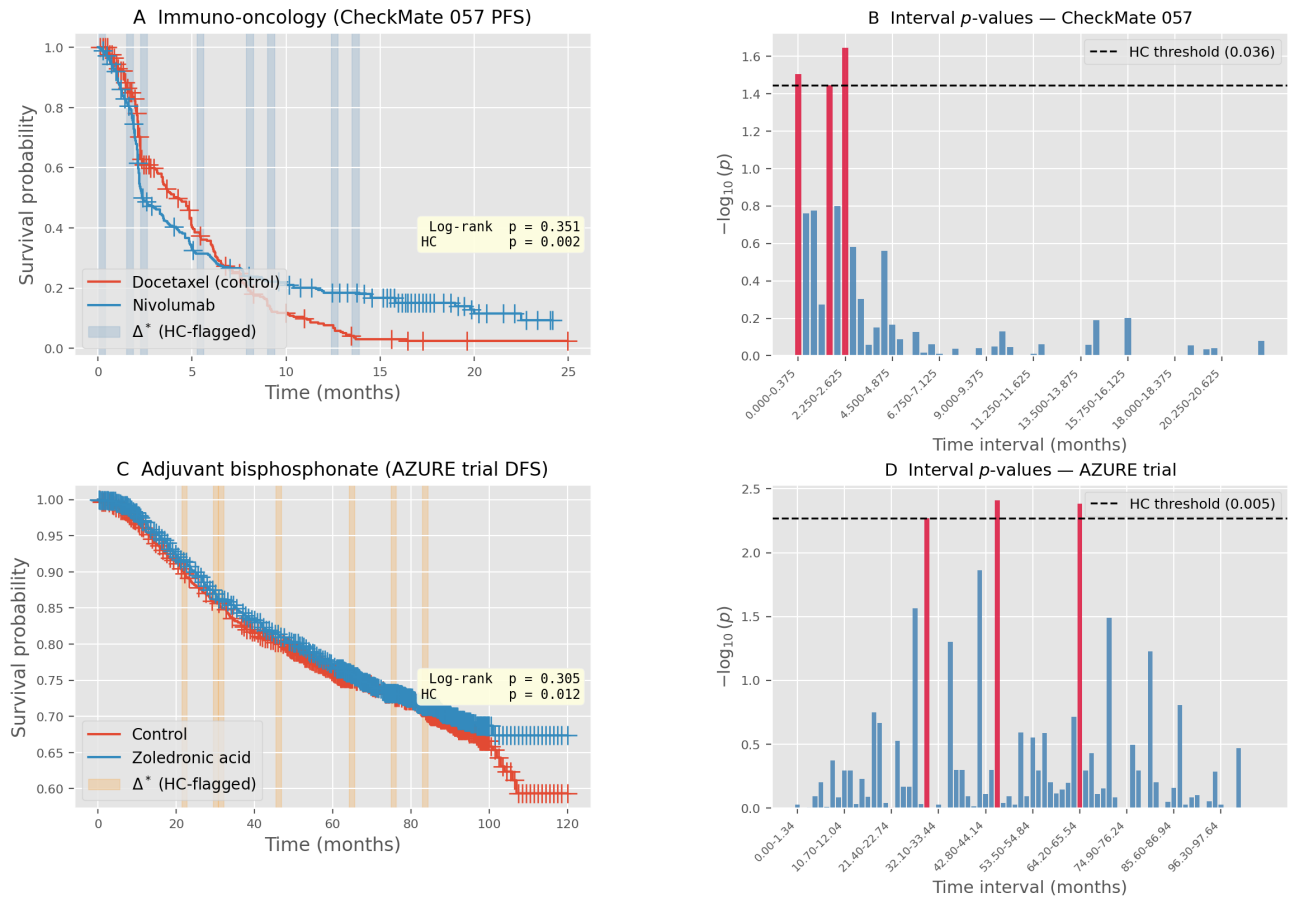


Fig. 1. Higher Criticism detects non-proportional hazard departures across two biological domains. (A) Kaplan-Meier progression-free survival curves for CheckMate 057 (nivolumab vs. docetaxel, $n = 582$) with HC-flagged intervals shaded in blue; log-rank $p = 0.351$, HC $p = 0.002$. Curves cross at ~ 3 months before diverging. (B) Per-interval $-\log_{10}(\text{hypergeometric } p\text{-value})$ for CheckMate 057. Bars exceeding the HC threshold (dashed, permutation-calibrated) are highlighted in red, marking the early-crossing and late-separation windows. (C) Kaplan-Meier disease-free survival for the AZURE trial (zoledronic acid vs. control, $n = 3359$) with HC-flagged intervals in orange; log-rank $p = 0.305$, HC $p = 0.012$. The signal reflects a menopause-dependent benefit accumulating in months 20–60. (D) Per-interval $-\log_{10}(\text{hypergeometric } p\text{-value})$ for AZURE. The signal is distributed across the mid-follow-up range, outside the reach of any single-weight log-rank alternative. Individual patient data reconstructed via Guyot et al. [2012].